

Swami Agnivesh Shaga

GitHub: github.com/agnivesh13

LinkedIn: [agniveshshaga](https://www.linkedin.com/in/agniveshshaga)

Portfolio: agnivesh.me

Email: agniveshshaga@gmail.com

Mobile: +91-8639871199

Location: Hyderabad, India

EDUCATION

- Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering & Technology** Hyderabad, India
Bachelor of Technology - Computer Science and Engineering 2027 (Expected)
Specialization: Artificial Intelligence & Machine Learning - Current CGPA: 8.55
- Acumen Junior College** Warangal, India
Intermediate (MPC) - Percentage: 91.6 2022
- Shine High School** Warangal, India
10th Class - CGPA: 10 2020

SKILLS SUMMARY

Programming: C, C++, R, Java, Python **LeetCode:** 1595 (link)

Data and Dev Tools: Relational Databases (MySQL), Docker, Git, GitHub Actions, **CodeChef:** 1567 (2 star) (link)

Flask, Agile/CI/CD

Cloud Technologies & Networking: AWS, pfSense, OpenVPN, Linux, VLANs, **Codeforces:** 1157 (Newbie) (link)

TCP/IP

EXPERIENCE

- Alfago Research LLP** Hyderabad, India
Quantitative Research Intern Dec 2025 – May 2026
 - Algorithm Engineering & Development:** Engineered automated systematic long/short trading algorithms for Silver and Natural Gas commodities, processing over 2 years of historical market data to achieve high-probability win rates of 88% and 83% respectively, yielding a simulated Rs. 22.34 Lakh return.
 - Statistical Testing & Validation:** Designed robust algorithmic validation frameworks utilizing Monte Carlo permutation tests, in-sample, and walk-forward analysis to ensure system resilience, prevent curve-fitting, and validate predictive models.
 - Data Analysis & Parameter Optimization:** Analyzed large financial datasets to dynamically optimize volatility-based parameters—including ATR-driven Stop Loss (SL) / Trailing Stop Loss (TSL), TEMA, and RSI—significantly maximizing risk-adjusted performance and system stability.
 - Financial Modeling Workflows:** Streamlined data analytics workflows for investment research by developing Discounted Cash Flow (DCF) models to evaluate complex equity valuations.

PROJECTS

- Automated OHLCV Data Pipeline for Algorithmic Trading** Feb 2025
Python (Asyncio, FastAPI), AWS (EC2, S3, Lambda), Parquet
 - Scalable Extraction Engine:** Orchestrated a highly resilient pipeline using AWS EC2 cron jobs to fetch 1-minute OHLCV data for all NSE-listed companies, utilizing AWS Lambda for secure, autonomous access token rotation.
 - High-Concurrency Engineering:** Engineered an asynchronous Python implementation of the Token Bucket algorithm, eliminating API rate-limit errors and maximizing throughput during massive data retrieval operations.
 - Big Data Optimization:** Streamlined Big Data ETL workflows to convert raw JSON logs into compressed Parquet files stored in S3, optimizing storage costs and analytical query performance.
 - FastAPI Retrieval Layer:** Developed a lightweight FastAPI microservice to provide low-latency, programmatic access to historical S3 datasets for downstream backtesting engines.
- Niyantrana – AI-driven NAFLD Risk Prediction System** Sep 2025
TensorFlow, Scikit-Learn, Flask, Docker, GitHub Actions
 - Predictive Modeling:** Built an LSTM-MLP fusion deep learning system achieving 88% ROC-AUC for early NAFLD prediction, forecasting biomarkers, and computing Fatty Liver Index scores.
 - RAG-based Coaching:** Integrated a RAG module using contextual health data to automate evidence-based lifestyle recommendations, reducing manual toil by 70%.
 - MLOps & CI/CD:** Implemented an Agile CI/CD workflow (GitHub Actions, Docker) for automated testing, model retraining, and resilient containerized Flask deployment.
- Enterprise-Grade Homelab Network Infrastructure** Dec 2025 – Jan 2026
pfSense, OpenVPN, VLANs, PKI, Unbound DNS, Omada SDN
 - Zero-Trust Architecture:** Designed a segmented network topology using 802.1Q VLANs to isolate critical assets, enforcing strict firewall rules to prevent lateral movement and enhance system security.
 - Secure Remote Access & Troubleshooting:** Deployed a custom OpenVPN server with an internal CA; actively troubleshooted and optimized mobile connectivity with custom keepalive tuning to mitigate carrier NAT timeouts.
 - Network Security & Privacy:** Implemented network-wide ad-blocking and IP reputation filtering; configured NAT redirection to force secure DNS resolution.

ACHIEVEMENTS

- LeetCode Coding Streak:** Maintained a 1-Year daily streak, solving 450+ problems in algorithms and data structures.
- Honourable Mention at CodeVerse (Turing Hut):** Secured a Top 10 rank among 300+ teams, successfully qualifying as one of the top 25 finalists.
- Independent Algo Trader:** Managed a Rs. 3.4 Lakh equity and commodities portfolio, applying disciplined risk management and quantitative strategies.
- SEBI Investor Certification:** Scored 50/50 in SEBI Investor Exam, demonstrating strong knowledge in equity, mutual funds, and financial markets.